MedTrack

Hospital Operations & Patient Analytics Dashboard

DASHBOARD STORYBOARD

Planning document for the four-page interactive dashboard implemented with Python, Streamlit, Pandas and Plotly. It documents page layouts, KPIs, visualizations, filters, navigation and important data limitations.

Dashboard pages	Hospital Overview · Patient Flow · Department Analytics · Resource Utilization
Application	Streamlit web app with interactive Plotly charts
Data note	Department, transfer and resource fields in the demo dataset are simulated
Purpose	Explore admissions, length of stay, billing summaries, patient flow and illustrative resource metrics

Scope: This storyboard describes the current Streamlit + Plotly implementation. It is not a Tableau workbook and does not generate a `.twbx` file.

1. Global Layout & Interaction Design

The app uses a sidebar for page selection and shared filters. The main content area presents KPI cards first, followed by charts and, where available, a summary table. Filters should update all compatible metrics and charts on the selected page.

MEDTRACK — HOSPITAL OPERATIONS & PATIENT ANALYTICS
SIDEBAR: Page selector | Admission Year | Gender | Medical Condition
KPI CARD: Total Admissions • Average Stay • Total Billing • Average Billing
MAIN AREA: Page-specific Plotly visualizations — trends, distributions, comparisons and resource charts
SUPPORTING AREA: Secondary chart(s), interpretation notes and/or summary table

Shared filters and navigation

Control	Purpose	Expected behavior
Dashboard selector	Navigate among the four pages	Opens the selected dashboard page
Admission Year	Focus on a chosen admission year	Filters compatible metrics and charts
Gender	Compare selected gender categories	Updates compatible charts and KPI values
Medical Condition	Focus on a selected condition	Updates compatible charts and KPI values

Navigation note: The current implementation uses a sidebar page selector; this storyboard does not claim Tableau-style parameter actions.

2. Dashboard 1 — Hospital Overview

Goal: provide a quick summary of hospital admissions and high-level billing and length-of-stay indicators.

KPI / visualization	Design	Question answered
Total Admissions	KPI card	How many admission records match the filters?
Average Stay	KPI card	What is the average recorded length of stay?
Total Billing	KPI card; currency unconfirmed	What is the sum of recorded billing amounts?
Average Billing	KPI card; currency unconfirmed	What is the average recorded billing amount?
Monthly Admission Trend	Line chart	How does admission volume change month by month?
Admissions by Type	Pie chart	What share belongs to each admission type?
Admissions by Medical Condition	Doughnut chart	How are records distributed by condition?

Layout sketch

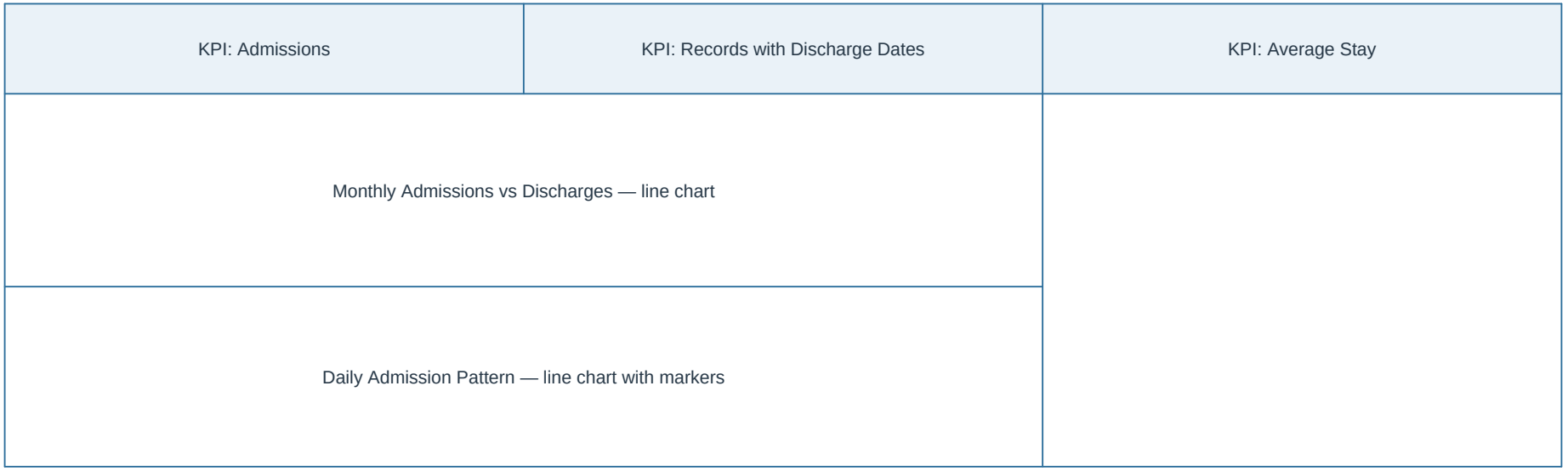
TOTAL ADMISSIONS	AVERAGE STAY	TOTAL BILLING	AVERAGE BILLING
Monthly Admission Trend — line chart			
Admissions by Medical Condition — doughnut chart			

Interpretation: Billing is shown without a currency symbol because the source currency is not confirmed.

3. Dashboard 2 — Patient Flow

Goal: understand admission and discharge patterns, length-of-stay variation and illustrative transfer patterns.

KPI / visualization	Design	Question answered
Admissions	KPI card	How many admission records match the filters?
Records with Discharge Dates	KPI card	How many records contain a discharge date?
Average Stay	KPI card	What is the average recorded stay duration?
Admissions vs Discharges	Monthly line chart	How do admissions and recorded discharges trend over time?
Length of Stay by Admission Type	Box plot	How does stay duration vary across admission types?
Daily Admission Pattern	Line chart with markers	Which dates have higher or lower admission counts?
Transfer Summary / Destinations	KPI + pie/doughnut; simulated	What illustrative transfer patterns appear in demo data?



Data caveat: The source dataset does not contain verified transfer records. Transfer counts and destinations are simulated and must not be presented as actual hospital activity.

4. Dashboard 3 — Department Analytics

Goal: compare patient volume and selected performance indicators across demo department categories.

KPI / visualization	Design	Question answered
Admission Share by Demo Department	Doughnut chart	How are records distributed across demo categories?
Average Stay by Demo Department	Bar chart, sorted by average stay	Which demo categories have higher average stay?
Average Stay vs Average Billing	Scatter plot	How do average stay and billing compare across demo categories?
Department Summary	Summary table	What are the aggregated values for each demo category?

WARNING: Department assignments are simulated demo categories
Admission Share by Demo Department — doughnut chart
Average Length of Stay by Demo Department — bar chart
Average Stay vs Average Billing — scatter plot
Department Summary Table

Data caveat: The original data does not provide verified department assignments. Department comparisons are illustrative and should be labelled demo/simulated.

5. Dashboard 4 — Resource Utilization

Goal: demonstrate resource-related views for capacity and workload monitoring while distinguishing simulated values from verified operational data.

KPI / visualization	Design	Question answered
Demo Bed Occupancy	KPI card / percentage	What occupancy percentage is represented by demo fields?
Average Demo Staff Count	KPI card	What is the average simulated staff value in filtered records?
Average Demo Equipment	KPI card	What is the average simulated equipment availability?
Monthly Admission Workload	Area chart	How does admission workload change month to month?
Admission Heatmap	Heatmap across months	How is admission volume distributed across the month dimension?
Admissions vs Estimated Stay Days	Scatter plot	How do monthly admissions relate to estimated total stay-days?
Average Simulated Staff by Month	Line chart	How do average demo staff values change by month?
Average Simulated Equipment Usage Hours by Month	Line chart	How do average demo usage hours change by month?

SIMULATED RESOURCE VALUES — DEMO ONLY
KPI: Demo Occupancy | Average Demo Staff | Average Demo Equipment
Monthly Admission Workload — area chart | Admission Heatmap — heatmap
Admissions vs Estimated Stay Days — scatter | Average Staff / Equipment Usage by Month — line charts

Data caveat: The source dataset does not contain verified daily bed census, staff rosters, equipment inventories or equipment-use logs. These fields were added for demonstration and must be labelled simulated.

6. KPI Definitions & Data Limitations

Interpret KPIs using the prepared fields and calculations in the code. Where source data is unavailable, a metric is not a verified hospital operational measure.

Metric	Meaning in this dashboard	Important limitation
Total Admissions	Count of filtered admission records	Do not assume this is a unique-patient count.
Average Length of Stay	Mean of the prepared length-of-stay KPI field	Interpretation depends on how that field was calculated and discharge dates are represented.
Total / Average Billing	Sum or mean of recorded billing amount	Currency label is not confirmed.
Demo Occupancy	Ratio from demo available/occupied-bed fields	Not a verified daily hospital bed census.
Average Demo Staff / Equipment	Mean of simulated values in filtered records	Not a confirmed hospital-wide roster or inventory.
Estimated Stay Days	Monthly total estimated from recorded length-of-stay values	Large totals (e.g. thousands of days) are totals, not average stay.
Readmission Rate	Not included as a verified KPI in the current dashboard	Reliable patient identifiers and longitudinal admission history are unavailable.
Department Metrics	Grouped by demo department labels	Original data does not provide verified department mapping.
Transfers	Simulated transfer counts/destinations	Not actual transfer records.

Recommended validation before submission

- Confirm KPI formulas against generate_hospital_kpis.py and the prepared dataset.
- Label simulated fields clearly in the app and report.
- Test page navigation, filters and chart titles.
- Document mentor approval for the change from Tableau/.twbx to Streamlit + Plotly.

7. Navigation & Project Delivery

Suggested workflow for a dashboard user

Step	Instructions
1. Open the app	Run the Streamlit app from the repository root.
2. Choose a page	Use the sidebar selector to open one of the four dashboard pages.
3. Apply filters	Select admission year, gender and medical condition as needed.
4. Explore charts	Hover over Plotly charts for tooltips and use available chart interactions.
5. Interpret carefully	Separate dataset-derived admission/billing/stay measures from simulated resource, department and transfer fields.

Suggested project files

Path / file	Purpose
scripts/hospital_dashboard_app.py	Streamlit app and Plotly visualizations
scripts/add_demo_resource_columns.py	Creates illustrative resource, department and transfer fields
data/hospital_dashboard_demo_data.csv	Dashboard dataset containing simulated demo fields
docs/dashboard_storyboard.pdf	This dashboard layout and interaction planning document
README.md	Setup instructions, dependencies, limitations and run command

Run command

```
python -m streamlit run scripts/hospital_dashboard_app.py
```

Final note: The original project specification names Tableau and .twbx files. This storyboard documents the current Streamlit + Plotly implementation and does not substitute for a Tableau workbook if the mentor requires one.